Term Paper final version

Depression is one of the most prevalent, debilitating, and potentially life-threatening worldwide. More than 300 million people worldwide suffer from depression, and they are often resistant to conventional treatments such as placebos and psychotherapy. (Stringaris, 2017). In recent years, however, psychedelic drugs such as ayahuasca have emerged as promising fast-onset antidepressants in treatment-resistant depression (Palhano-Fontes et al., 2018). These psychedelic drugs have shown potential therapeutic efficacy in many psychiatric disorders due to their effects on brain function, primarily acting on serotonin 5-HT2A receptors, which can help alter the states of perception and emotion. (Aznar & Klein, 2013). This led researchers to consider whether psychedelic drugs can be an effective treatment for depression. In this paper, I argue that psychedelic drugs can be an effective treatment for major depressive disorder in both the short and long term. This argument is supported by the fact that psychedelic drugs induce rapid and high remission rates, reduce psychological flexibility, and provide lasting antidepressant effects that ensure a long-term reduction of depressive symptoms. Understanding the potential therapeutic effects of psychedelic drugs is essential for developing more effective treatments for depression.

Psychedelic drugs like ayahuasca produce high remission rates, which lead to a rapid reduction in depression severity. Ayahuasca is a psychoactive ethnobotanical that contains dimethyltryptamine (DMT), an endogenous compound that acts on serotonergic 2A receptors and contributes to the increases in introspection and

serenity that help reduce depression. The reaction to a single ayahuasca dose occurs rapidly. The mechanism behind this antidepressant effect is supported by neurophysiological evidence based on a SPECT study that shows ayahuasca increases bilateral activation of those brain areas that are involved in the modulation of emotions, including the anterior cingulate cortex, frontomedial cortex, and amygdala. Furthermore, resting-state fMRI studies have demonstrated that ayahuasca decreases activation within key regions of the brain's default mode network and then reduces depression by modulating brain structures (Santos et al., 2017). Palhano-Fontes and his research group conducted a double-blind parallel-arm randomized placebo-controlled trial, which separated 29 adults aged 18-60 years with unipolar major depressive disorder and currently in a moderate to severe depressive episode. They were randomly assigned to receive either ayahuasca or placebos. This study measured the severity of depression at baseline and at Days 1, 2, and 7 using both the Montgomery-Åsberg Depression Rating Scale (MADRS) and the Hamilton Depression Rating Scale (HAM-D). The data demonstrated that ayahuasca has significant antidepressant effect compared to placebo base on the statistically different response rate at Day7, when 36% of participants in the ayahuasca group met remission compared to 7% in the placebo group provided by MADRS and a significant trend in HAM-D remission rate at Day 7, when 43% participants in ayahuasca group met remission compared to 13% in placebo group (Palhano-Fontes et al., 2018). These findings indicate that ayahuasca can induce rapid and meaningful reductions in depression. Therefore, psychedelic drugs like ayahuasca can be an

effective short-term treatment for individuals who suffer from major depression.

Psychedelic-assisted therapy provides significant reductions in psychological flexibility, which limits individuals' mindfulness and values-congruent living and contributes to both rapid and sustained antidepressant effects. Psilocybin is a tryptamine alkaloid found in Psilocybe mushrooms (Goodwin et al., 2022). It can interact with serotonin 5-HT2A receptors in the brain and influence an individual's capacity of adapting internal and external demands by mediating openness, emotional processing, and self-perspective. Sloshower and his research group conducted a placebo-controlled, within-subject clinical trial to measure changes in psychological flexibility among adults aged 18-35 with major depressive disorder. Participants received placebos first and psilocybin later over a 16-week period, and their psychological flexibility will be assessed using the Acceptance and Action Questionnaire-II (AAQ-II). The data shows significant reductions in psychological inflexibility at 2 weeks post-psilocybin, along with increases in mindfulness domains at two- and four-weeks post-psilocybin relative to baseline and 2 weeks pre-psilocybin. These changes indicate that psilocybin has a great contribution to the reductions in psychological flexibility, which is strongly associated with reductions in depression severity (Sloshower et al., 2024). In separate research, a double-blind, randomized clinical trial created by Goodwin and his research group administered a dose of 25mg, 10mg, or 1mg psilocybin to randomly assigned adults who suffered from major depressive disorder and used MADRS scores to measure the severity of their depression for 12 weeks after treatment. The data shows a rapid and significant

decline in MARDS scores in the 25-mg psilocybin group maintained for 12 weeks and might be even longer compared to other groups (Goodwin et al., 2022). These two studies provide converging evidence that psilocybin-assisted therapy promotes lasting improvement in psychological flexibility, resulting in both immediate and sustained reduction in depressive symptoms.

The mild adverse effects that the psychedelic drugs have, including anxiety and transient headaches, are manageable in a controlled setting, which allows psychedelics to be generally well-tolerated (Andrade et al., 2025). This feature enables patients to receive repeated dosing, thereby producing lasting antidepressant effects and ensuring a long-term reduction of depressive symptoms. Ellis and his research group reported the long-term results of an open-label pilot study, which shows the effects of 25mg psilocybin in veterans who had major depressive episode and failed to respond to over 5 treatments or lasting over 2 years. Their depression scores will be measured by the MADRS scores across a 12-month period. The figure indicates that the total MADRS scores declined by over 50% from baseline to the end, with prominent decreases observed at Month 6, Month 9, and particularly at Month12, where the MADRS scores were 54.23% lower on average compared to baseline. The data demonstrate that depression scores for psychedelic drugs like psilocybin showed a robust and sustained reduction up to 12 months. (Ellis et al., 2025). The finding that the tolerability of psilocybin and the sustained reductions in depression observed across a full year support my reason that psychedelic drugs can provide reliable long-term relief when administered in a controlled setting and monitored appropriately.

Some individuals might argue that psychedelic drugs like psilocybin and ayahuasca that show promise as fast-acting antidepressants are not suitable for all populations. For some reasons, the antidepression effects of psilocybin will diminish faster than placebo and other conventional treatments. The figure in Carhart-Harris's trial shows that participators in the psilocybin group have worsened their symptoms around Day 14, which is much earlier than the conventional treatment using escitalopram, and the decreasing line of escitalopram is approximately linear (Carhart-Harris et al., 2021). This might indicate that the antidepressant effects in conventional treatments are more stable than psychedelic-assisted therapy.

However, although there are still some uncertainties in psychedelic-assisted therapy, this does not necessarily indicate a lack of stability or efficacy. During Carhart-Harris's six-week trial, 70% patients in the psilocybin group achieved remission, which was 20% more than the escitalopram group. Overall, psychedelic drugs provide higher remission for depression compared to conventional treatments. (Carhart-Harris et al., 2021). The conclusion should base on a complete trial, but not a single specific point in the trial. For those uncertainties such as adverse effects are manageable if you control each dose in a proper setting (Andrade et al., 2025).

In conclusion, current evidence demonstrates that psychedelic drugs have the capacity to rapidly reduce depressive severity and sustain antidepressant effects over a long period. The findings in this paper show that psychedelic-assisted therapy holds strong potential in both short- and long-term treatment for major depressive disorder. As the incidence of depression continues to rise globally, this safe and effective

treatment with psychedelics not only reduces individuals' depressive severity but also reduces the negative impact of depression on society.

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